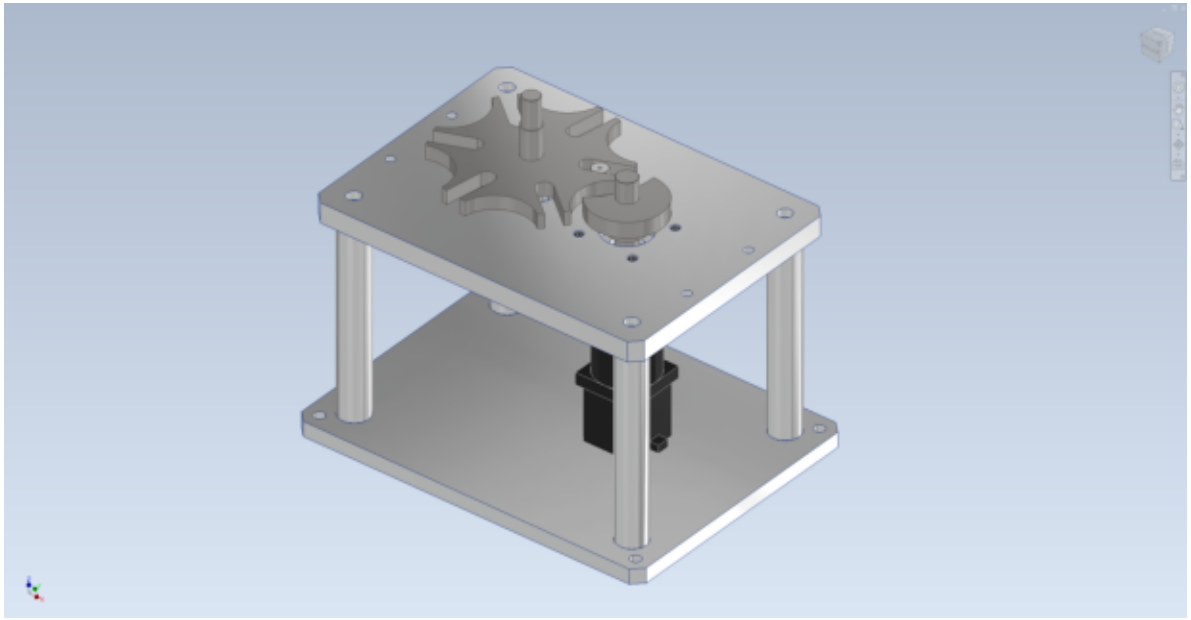
Geneva Indexing Table: 6 stations, 6-slot Geneva with locking disc

GenevaMechanism - calculation report. Designed and calculated only: nothing has been built or tested. CAD = measured in Inventor, CALC = calculated, CAT = catalogue / datasheet.

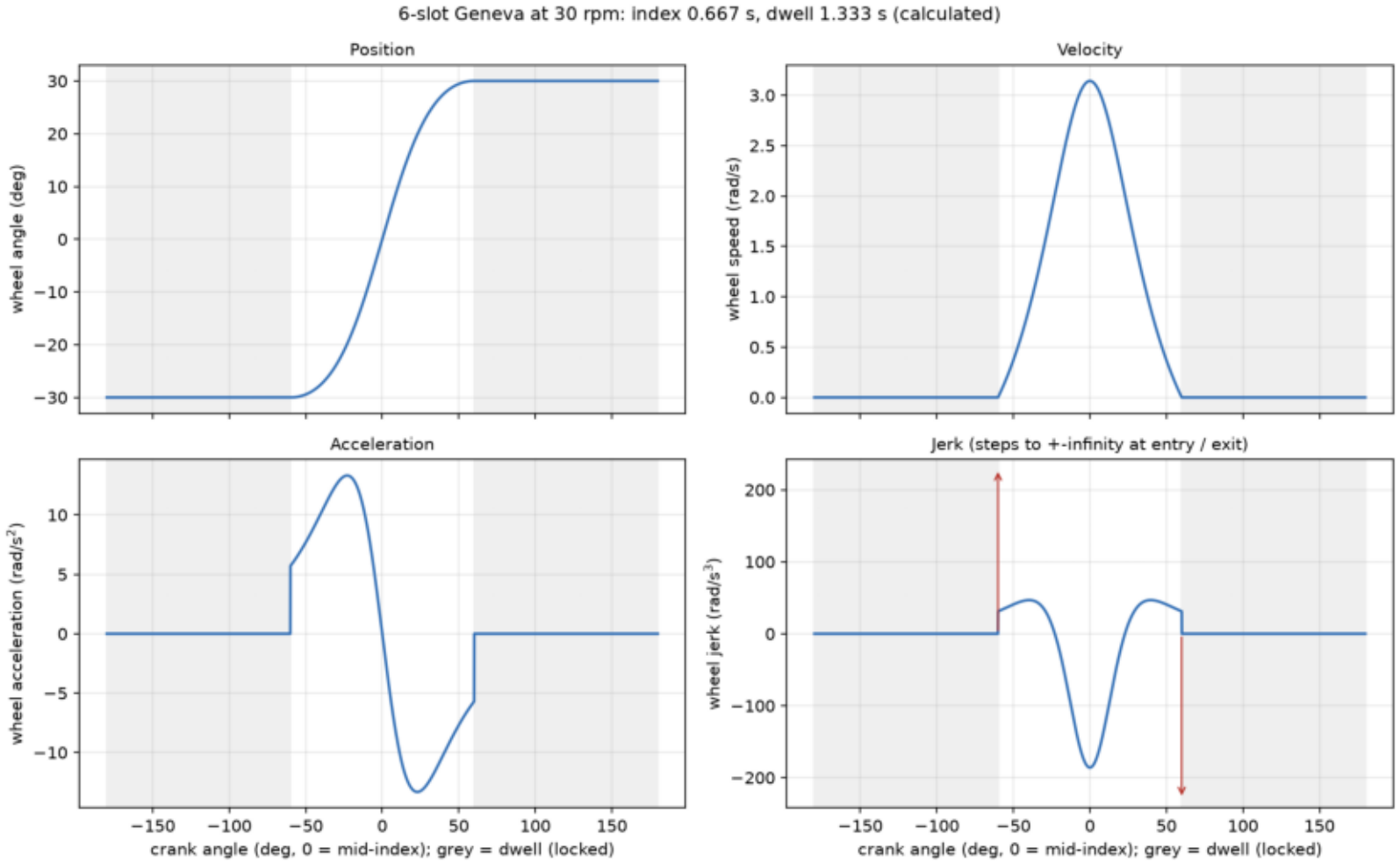


Item	Value	Source
Mechanism	external 6-slot Geneva, tangent entry: C 100 mm, crank a 50 mm (= C sin 30), wheel R87 x 14, INA KR16-PP roller, locking disc R36	CALC/CAT
Motion	60 deg per crank turn: 120 deg engaged, 240 deg dwell (dwell fraction 0.667)	CALC
Design point	30 rpm crank = 30 indexes/min: index 0.667 s, dwell 1.333 s	CALC
Drive	Oriental Motor PKP264D14A2-PLE60-5B (NEMA 23 stepper, Neugart PLE60 5:1), $502.50, Ruland MJC41 coupling	CAT
Output inertia	0.0793 kg m2 (0.0397 Inventor + 0.0397 payload 6 x 0.5 kg)	CAD/CALC
Peak torques	crank 0.87 Nm rigid / 0.89 Nm dynamic; wheel 1.11 Nm	CALC
Motor margin	SF 2.12 on the estimated pull-out torque (2.29 Nm) at 1.25 x rigid peak	CALC
Locking	0.06 deg wheel play in full dwell = 0.12 mm at the stations (R115)	CALC
Assembly	91 occurrences, 270 constraints, 0 DOF; 0 clashes at 28 crank angles; kinematics error 0.0 deg	CAD
Mass / envelope	23.2 kg; 370 x 300 x 359 mm	CAD

1. Geneva kinematics: position, velocity, acceleration, jerk

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Geometry: for an n-slot external Geneva the crank radius is chosen so the crank is perpendicular to the slot when the roller enters (tangent entry): a = C sin(pi/n) = 50 mm, lambda = a/C = 0.5. The crank is engaged for 2 alpha0 = 180 - 360/n = 120 deg and the wheel turns 360/n = 60 deg. With psi measured from mid-index: beta = atan(lambda sin psi / (1 - lambda cos psi)); beta' = w lambda (cos psi - lambda) / (1 - 2 lambda cos psi + lambda^2); beta'' = w^2 lambda sin psi (lambda^2 - 1) / (1 - 2 lambda cos psi + lambda^2)^2. At 30 rpm (w = 3.142 rad/s): peak wheel speed 3.142 rad/s (v/w = 1.00 at mid-index), peak acceleration 13.32 rad/s2 (1.350 w^2 at psi = 22.9 deg), acceleration step at entry 5.70 rad/s2 (0.577 w^2 = tan 30 deg), peak interior jerk 186.0 rad/s3. At the station radius (115 mm): 0.361 m/s and 1.53 m/s2 (0.16 g). The same law is written into Inventor as the parameter Wheel(Crank); the solved poses match design.py with 0.0 deg error at 28 crank angles (CAD).

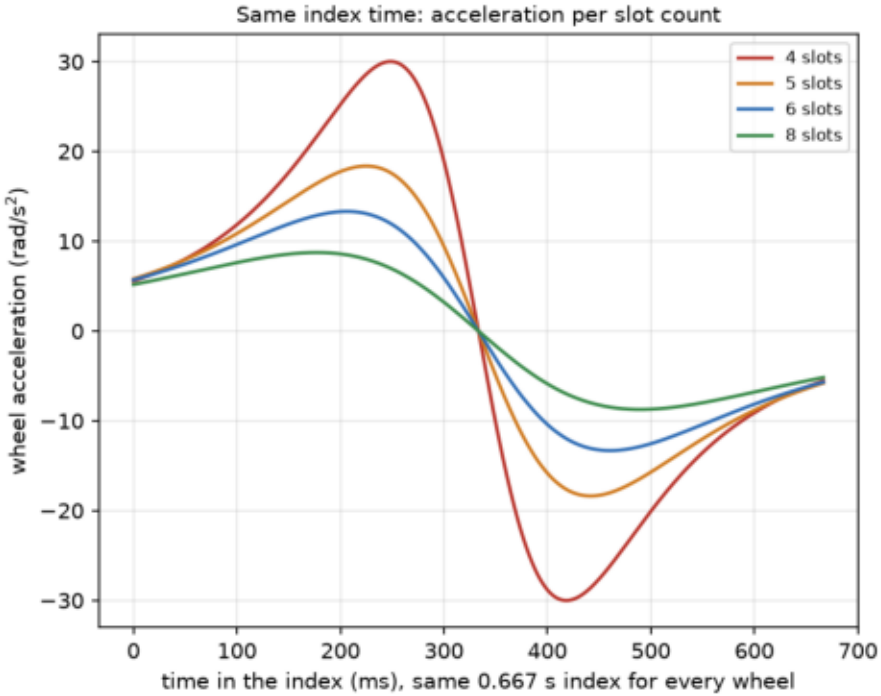


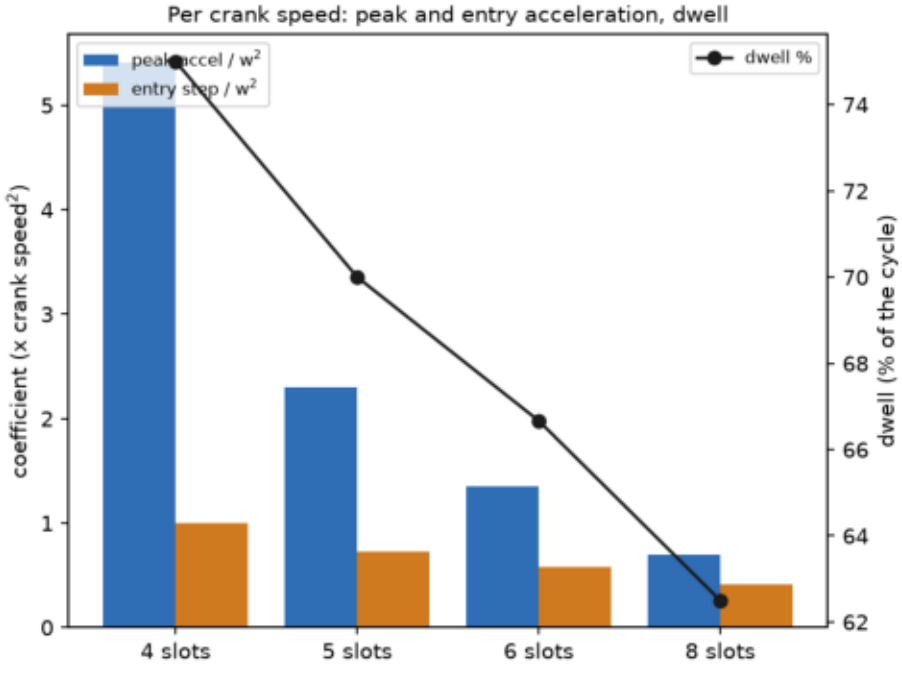
2. Why six slots: 4 / 5 / 6 / 8-slot comparison

GenevaMechanism - calculation report. Designed and calculated only: nothing has been built or tested. CAD = measured in Inventor, CALC = calculated, CAT = catalogue / datasheet.

Slots	Engaged (deg)	Dwell	v max / w	a max / w2	Entry step / w2	Crank torque / (J	rpm for 0.667 s i	a max at that rpm (rad
4	90	75.0 %	2.414	5.407	1.000	10.015	22.5	30.0
5	108	70.0 %	1.426	2.299	0.727	2.458	27.0	18.4
6	120	66.7 %	1.000	1.350	0.577	0.990	30.0	13.3
8	135	62.5 %	0.620	0.700	0.414	0.306	33.8	8.7

Decision: 6 slots. A 4-slot wheel has 4x the peak acceleration coefficient and 10x the crank-torque coefficient of a 6-slot one at the same crank speed, and an entry step of 1.0 w^2; 8 slots is gentler but gives only 45 deg steps, a 62.5 % dwell and a crank radius of 38 mm on a 100 mm centre distance (weaker roller arm, larger wheel for the same pitch). Six matches the six stations directly and keeps the crank torque coefficient below 1.0. Values are coefficients of the crank speed, from the closed-form law (CALC).

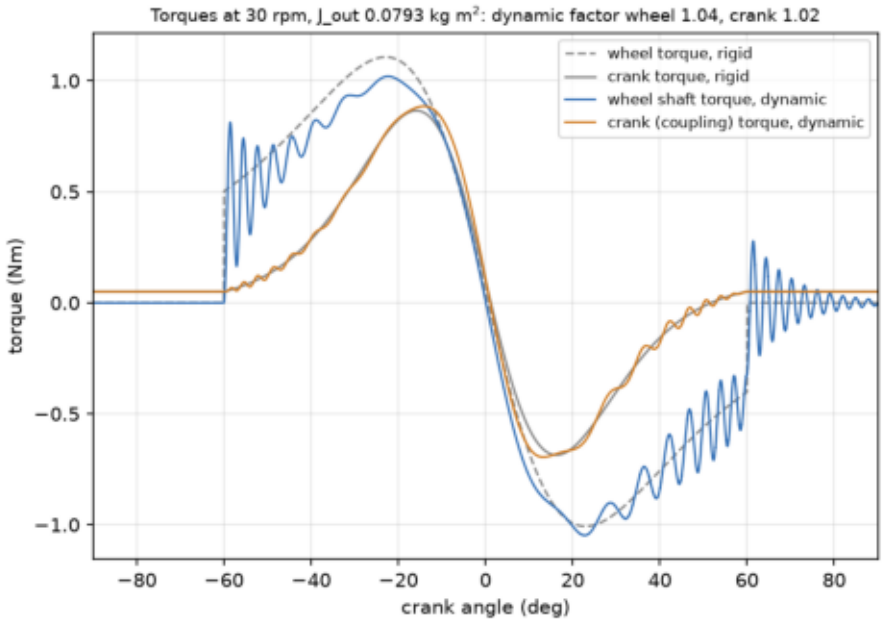


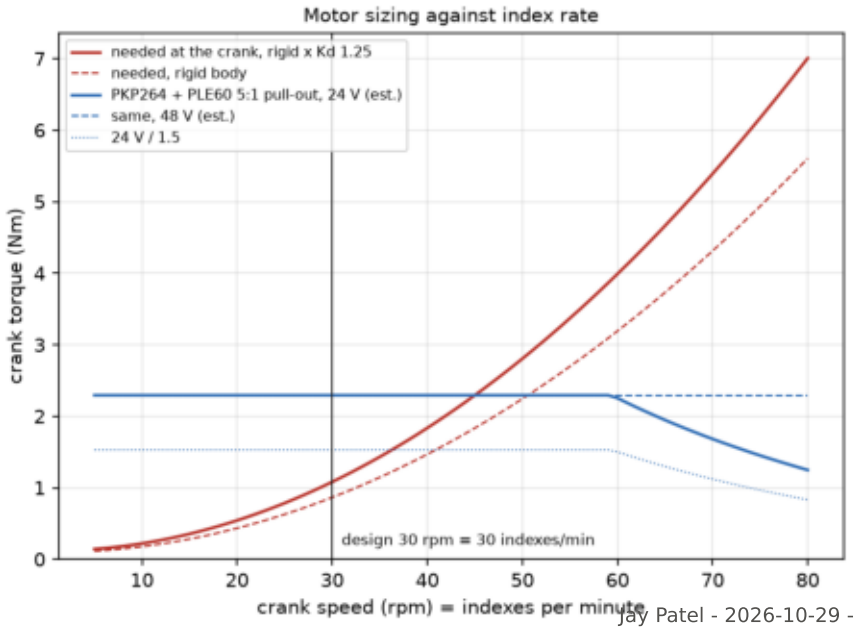


3. Torques, dynamic factor and motor sizing

GenevaMechanism - calculation report. Designed and calculated only: nothing has been built or tested. CAD = measured in Inventor, CALC = calculated, CAT = catalogue / datasheet.

Inertia (CAD): wheel group 0.0397 kg m2 about the wheel axis, summed by parallel-axis over the 91 occurrences (check: Inventor table 0.02562 vs disc formula 0.02552); payload 6 x 0.5 kg at 115 mm adds 0.0397; crank group 0.00050; motor rotor + gearbox reflected 0.00056 kg m2. Rigid body: T_wheel = J beta'' (+0.05 Nm seal drag), T_crank = T_wheel beta'/w + 0.05 Nm. Peak crank torque 0.866 Nm driving at -15.8 deg and -0.686 Nm braking (the wheel back-drives the crank while it decelerates); RMS 0.272 Nm, peak power 2.72 W, mean 0.18 W.
Dynamics: 3-DOF torsional model (stepper field -> rotor -> PLE60 + MJC41 -> crank -> Geneva -> wheel shaft -> table), stiffnesses stepper field 844, gearbox 6875 (Neugart), coupling 600 (assumed), wheel-to-table 11231 Nm/rad (half the calculated shaft value 22461, for keys and bolted joints), zeta 0.05. Table mode 61.4 Hz. The step in acceleration at entry rings the table at that frequency (left plot). Dynamic factor at the nominal stiffnesses: wheel 1.04, crank 1.02; over coupling 300-3000 Nm/rad and wheel-to-table x0.5-x2: wheel 1.01-1.94, crank 1.01-1.05. Design factors used: 2.0 on the wheel side, 1.25 on the crank / motor side. A soft coupling is what raises the wheel-side factor (crank speed ripple), so the stiffest 98A spider is specified.
Motor: PKP264D14A2 + PLE60 5:1, 464 oz-in (3.28 Nm) holding at the output (listing). Pull-out at 150 motor rpm estimated with an R-L-back-EMF model and a 0.7 running factor: 2.29 Nm at the output (Oriental's curve is an image I did not read). Need 1.08 Nm: SF 2.12 (2.59 on the simulated peak). Highest rate with SF 1.5: 36 indexes/min. Stepper load angle peaks at 0.60 electrical deg (stall at 90). Load inertia at the motor peaks at 225:1 mid-index and is zero in the dwell; it is handled as torque demand because the crank runs at constant commanded speed. The controller starts and stops the crank only inside the dwell, where it accelerates just the crank side (0.005 Nm), never the table.



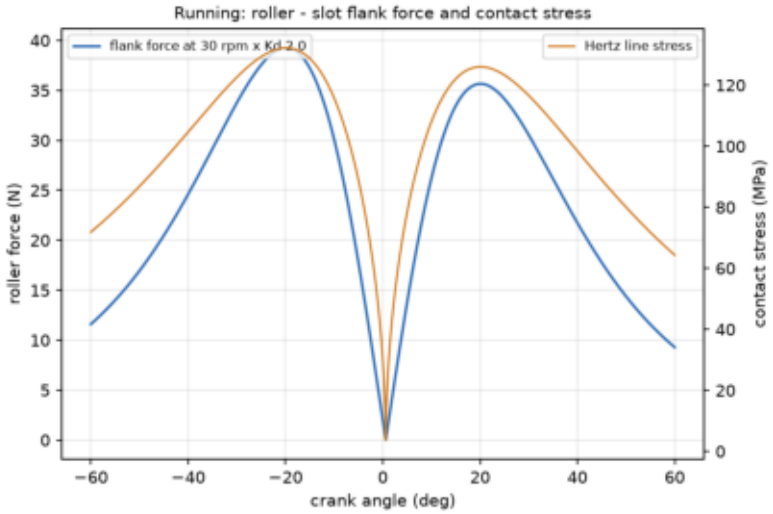


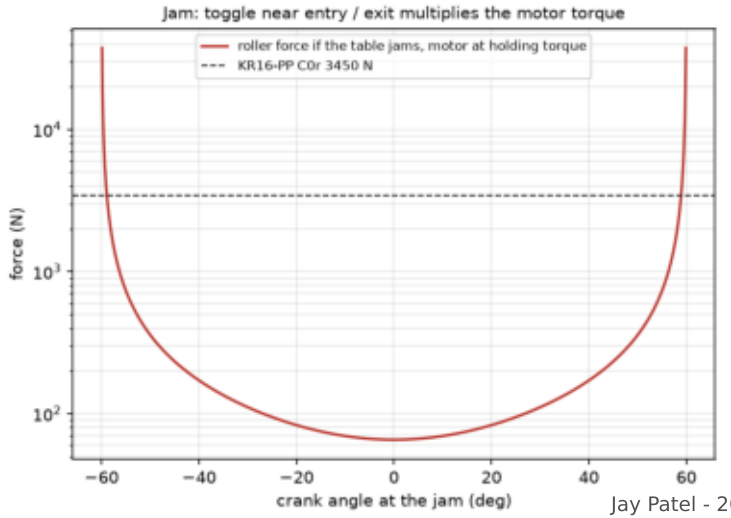
4. Roller / slot contact, jam loads, shafts, keys, bearings

GenevaMechanism - calculation report. Designed and calculated only: nothing has been built or tested. CAD = measured in Inventor, CALC = calculated, CAT = catalogue / datasheet.

Item	Result	Source
Flank force (running, x Kd 2.0)	39.2 N peak at psi -19.8 deg (F = T_wheel / r_roller)	CALC
Hertz line contact, roller 16 on flat flank	132 MPa over 10 mm; SF 11.3 on 1500 MPa (50 HRC flank); 1239 MPa even at the roller's C0r	CALC
KR16-PP roller (Cr 3200, C0r 3450, Cu 465 N)	static SF 88; below the fatigue limit load: True; 1.46 roller turns per index, L10 > 10^8 h	CALC/CAT
Jam (table blocked, motor at holding torque)	roller force 66 N at mid-index, rising without limit toward entry (toggle): above C0r in the first 1.0 deg, wheel torque above the 5 mm table key capacity (18.7 Nm) in the first 13.9 deg after entry	CALC
Shafts	wheel shaft 17 mm: 2.3 MPa running; crank shaft at motor holding torque 2.1 MPa; wheel shaft 310 MPa if a jam ever reached C0r (overload case)	CALC
Keys DIN 6885	driver key 16.4 MPa at holding torque, table key 11.8 MPa running (allow 100)	CALC
Roller stud 6 mm bending	12.4 MPa at the peak flank force	CALC
Bearings 4 x 6204-2RSH (C0 6.55 kN)	axial 82 N (table group + payload) on the lower wheel bearing, static SF 80, below Pu: True; 8 kg placed at the table rim makes 218 N per bearing (54 mm span)	CALC/CAT

Decision: the running loads are small (the drive is sized by the motor's speed-torque curve, not by strength), but a jam just after entry is dangerous: near entry the slot force has almost no moment arm about the crank, so the motor torque is multiplied without limit. A current limit does not protect it. Recommended: stall detection on the stepper (encoder or StallGuard-type driver) and a ball-detent torque limiter in the table hub, set near 5 Nm (about 2.5 x the dynamic peak), that re-engages every 60 deg.





5. Entry shock, locking accuracy and Inventor checks

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Entry shock: with tangent entry the roller meets the slot moving exactly along it (2D check: 0.0 deg between the roller velocity and the slot axis), so the wheel starts from zero speed: no velocity jump and no impact. What remains is a step in acceleration of 5.70 rad/s2 (a torque step of 0.45 Nm into the table), i.e. infinite jerk at entry and exit; it rings the table at its 61 Hz mode (decays to 6.1e-05 deg in the dwell). A radial slot with a non-tangent entry would hit the slot with a finite sideways speed instead. The geometry is forgiving: a crank radius 0.5 mm too long gives a 0.33 deg entry angle and a 0.0105 rad/s speed jump, 1.0 mm gives 0.0212 rad/s (0.7 % of the peak speed); the drawing still holds a 50.00 +/-0.01 and C 100.00 +/-0.01 so the entry stays clean. At mid-index the acceleration changes sign and the roller crosses the 0.06 mm slot clearance to the other flank in 34 ms, closing at 5.3 mm/s (a light tap; it is why the slot is held to +0.04/+0.08).
Locking: the disc (R36) runs in R36.10 arcs on the wheel. 2D check (shapely, outline of the CAD parts): wheel play 0.06 deg from 120 deg of crank onward (0.12 mm at R115), 0.12 deg at 90 deg, but 0.94 deg (1.9 mm) just after exit, because the relief that lets the wheel tips pass also opens one horn of the lock. Work at the stations should happen between crank 90 and 270 deg (180 deg = 1.0 s of the 1.33 s dwell); a shot pin is the upgrade for better than +/-0.1 mm.

Check	Result	Source
DOF / constraints	0 DOF, 270 constraints {'Flush': 233, 'Mate': 35, 'Angle': 2}, 0 unhealthy	CAD
Interference	0 clashing pairs at 28 crank angles from -180 to 420 deg (dense through +-60); a deliberate 2 deg wheel offset produced clashes (156.7 mm3 roller / slot), so the check is live	CAD
Moving clearances (min over the sweep)	wheel - locking disc 0.10 mm, wheel - roller 0.03 mm (slot clearance), wheel - crank shaft 3.03 mm, wheel - crank spacer 1.03 mm	CAD
2D outline check	disc gap while indexing 0.100 mm, roller gap 0.030 mm, shaft to wheel tip 3.00 mm	CALC
Drawings	GA: overall [370.0, 359.0] + [300.0] kept, centre distance 100.00 checked, 33 parts-list rows; manufacturing: 4 sheets, every auto dim checked, wrong ones deleted	CAD
Mass	total 23.20 kg: frame 14.62, wheel + table 5.31, gearmotor 1.65, crank group 0.87 kg	CAD

6. Assumptions, sources and limits

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Catalogue data (read online 2026-10-29): Oriental Motor PKP264D14A2-PLE60-5B catalogue page (464 oz-in holding, 1.4 A, 2.1 ohm, 6 mH, 0.77 oz-in2 rotor, 5:1, 10 arcmin, 14 mm shaft, total 4.44 in, web price $502.50); Neugart PLE catalogue 05/2026 (PLE060 1-stage: T2N 40 Nm at i = 5, T2max 61 Nm, 2.0-2.7 Nm/arcmin, 0.085-0.149 kg cm2 input inertia, 700 N radial, 0.9 kg, 52 PCD 4 x M5 x 8, 40 h7 pilot, 14 h7 x 35 shaft, housing 47 mm); Schaeffler KR16-PP (D 16, C 11, B 28, B1 12.2, B2 16, M6x1, Cr 3200 N, C0r 3450 N, Cu 465 N); Ruland MJC41 (41.3 OD, 53 mm; hub prices from RS: MJC41-14-A GBP 26.71, MJC41-20-A HKD 248.80); SKF 6204-2RSH (13.5 / 6.55 kN, class values).
Assumed: MJC41 98A torsional stiffness 600 Nm/rad (Ruland's figure was not in the listings I could read; a 300-3000 range was simulated); wheel-to-table joint stiffness = half the shaft value; damping ratio 0.05; seal drag 0.025 Nm per 6204-2RSH; motor pull-out torque from an R-L model with a 0.7 running factor instead of Oriental's published curve; motor / adapter split inside the 4.44 in length and 1.65 kg mass (envelope); payload 0.5 kg per station at 115 mm; contact allowable 1500 MPa for induction-hardened 4140 flanks; key crushing allowable 100 MPa.
Not checked: the stepper's mid-band resonance near 2.5 rev/s (150 motor rpm) - run 1/16 microstepping or finer; plate deflection and line-boring tolerance build-up (100.00 +/-0.01 is a drawing requirement); thermal growth; noise; grease life on the slot flanks. The driver, supply, home sensor, controller and guard are in the BOM but not modelled.
Cost: about 1811*forthepartsandestimatedmachining*(568 of it confirmed on vendor pages: gearmotor and coupling hubs), see GenevaMechanism_BOM.csv and Tests\Buildability\buildability_check.md.
Files: Analysis\geneva_analysis.py (all calculations), Tests\Kinematics\geneva2d.py (outline check), _build*.py (Inventor model, checks, drawings).